

OpenStax Calculus Volume 2

Section 3.4 Exercises: Partial Fractions

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Express the rational function as a sum or difference of two simpler rational expressions.

1. $\frac{1}{(x-3)(x-2)}$
2. $\frac{x^2+1}{x(x+1)(x+2)}$
3. $\frac{1}{x^3-x}$
4. $\frac{3x+1}{x^2}$
5. $\frac{3x^2}{x^2+1}$ (Hint: Use long division first.)
6. $\frac{2x^4}{x^2-2x}$
7. $\frac{1}{(x-1)(x^2+1)}$
8. $\frac{1}{x^2(x-1)}$
9. $\frac{x}{x^2-4}$
10. $\frac{1}{x(x-1)(x-2)(x-3)}$
11. $\frac{1}{x^4-1} = \frac{1}{(x+1)(x-1)(x^2+1)}$
12. $\frac{3x^2}{x^3-1} = \frac{3x^2}{(x-1)(x^2+x+1)}$
13. $\frac{2x}{(x+2)^2}$
14. $\frac{3x^4+x^3+20x^2+3x+31}{(x+1)(x^2+4)^2}$

Use the method of partial fractions to evaluate each of the following integrals.

15. $\int \frac{dx}{(x-3)(x-2)}$
16. $\int \frac{3x}{x^2+2x-8} dx$
17. $\int \frac{dx}{x^3-x}$
18. $\int \frac{x}{x^2-4} dx$

19. $\int \frac{dx}{x(x-1)(x-2)(x-3)}$
20. $\int \frac{2x^2+4x+22}{x^2+2x+10} dx$
21. $\int \frac{dx}{x^2-5x+6}$
22. $\int \frac{2-x}{x^2+x} dx$
23. $\int \frac{2}{x^2-x-6} dx$
24. $\int \frac{dx}{x^3-2x^2-4x+8}$
25. $\int \frac{dx}{x^4-10x^2+9}$

Evaluate the following integrals, which have irreducible quadratic factors.

26. $\int \frac{2}{(x-4)(x^2+2x+6)} dx$
27. $\int \frac{x^2}{x^3-x^2+4x-4} dx$
28. $\int \frac{x^3+6x^2+3x+6}{x^3+2x^2} dx$
29. $\int \frac{x}{(x-1)(x^2+2x+2)^2} dx$

Use the method of partial fractions to evaluate the following integrals.

30. $\int \frac{3x+4}{(x^2+4)(3-x)} dx$
31. $\int \frac{2}{(x+2)^2(2-x)} dx$
32. $\int \frac{3x+4}{x^3-2x-4} dx$ (Hint: Use the rational root theorem.)

Use substitution to convert the integrals to integrals of rational functions. Then use partial fractions to evaluate the integrals.

33. $\int_0^1 \frac{e^x}{36-e^{2x}} dx$ (Give the exact answer and the decimal equivalent. Round to five decimal places.)
34. $\int \frac{e^x}{e^{2x}-e^x} dx$
35. $\int \frac{\sin x dx}{1-\cos^2 x}$
36. $\int \frac{\sin x}{\cos^2 x + \cos x - 6} dx$
37. $\int \frac{1-\sqrt{x}}{1+\sqrt{x}} dx$
38. $\int \frac{dt}{(e^t - e^{-t})^2}$
39. $\int \frac{1+e^x}{1-e^x} dx$
40. $\int \frac{dx}{1+\sqrt{x+1}}$
41. $\int \frac{dx}{\sqrt{x} + \sqrt[4]{x}}$
42. $\int \frac{\cos x}{\sin x(1-\sin x)} dx$

43. $\int \frac{e^x}{(e^{2x}-4)^2} dx$

44. $\int_1^2 \frac{1}{x^2\sqrt{4-x^2}} dx$

45. $\int \frac{1}{2+e^{-x}} dx$

46. $\int \frac{1}{1+e^x} dx$

Use the given substitution to convert the integral to an integral of a rational function, then evaluate.

47. $\int \frac{1}{t-\sqrt[3]{t}} dt$; $t = x^3$

48. $\int \frac{1}{\sqrt{x+\sqrt[3]{x}}} dx$; $x = u^6$

49. Graph the curve $y = \frac{x}{1+x}$ over the interval $[0, 5]$. Then, find the area of the region bounded by the curve, the x -axis, and the line $x = 4$.

[Figure: This figure is a graph of the function $y = x/(1+x)$. The graph is only in the first quadrant. It begins at the origin and increases into the first quadrant. The curve stops at $x = 5$.]

50. Find the volume of the solid generated when the region bounded by $y = 1/\sqrt{x(3-x)}$, $y = 0$, $x = 1$, and $x = 2$ is revolved about the x -axis.

51. The velocity of a particle moving along a line is a function of time given by $v(t) = \frac{88t^2}{t^2+1}$. Find the distance that the particle has traveled after $t = 5$ sec.

Solve the initial-value problem for x as a function of t .

52. $(t^2 - 7t + 12)\frac{dx}{dt} = 1$, $t > 4$, $x(5) = 0$

53. $(t + 5)\frac{dx}{dt} = x^2 + 1$, $t > -5$, $x(1) = \tan 1$

54. $(2t^3 - 2t^2 + t - 1)\frac{dx}{dt} = 3$, $x(2) = 0$

55. Find the x -coordinate of the centroid of the area bounded by $y(x^2 - 9) = 1$, $y = 0$, $x = 4$, and $x = 5$. (Round the answer to two decimal places.)

56. Find the volume generated by revolving the area bounded by $y = \frac{1}{x^3+7x^2+6x}$, $x = 1$, $x = 7$, and $y = 0$ about the y -axis.

57. Find the area bounded by $y = \frac{x-12}{x^2-8x-20}$, $y = 0$, $x = 2$, and $x = 4$. (Round the answer to the nearest hundredth.)

58. Evaluate the integral $\int \frac{dx}{x^3+1}$.

For the following problems, use the substitutions $\tan(\frac{x}{2}) = t$, $dx = \frac{2}{1+t^2}dt$, $\sin x = \frac{2t}{1+t^2}$, and $\cos x = \frac{1-t^2}{1+t^2}$.

59. $\int \frac{dx}{3-5\sin x}$

60. Find the area under the curve $y = \frac{1}{1+\sin x}$ between $x = 0$ and $x = \pi$. (Assume the dimensions are in inches.)

61. Given $\tan(\frac{x}{2}) = t$, derive the formulas $dx = \frac{2}{1+t^2}dt$, $\sin x = \frac{2t}{1+t^2}$, and $\cos x = \frac{1-t^2}{1+t^2}$.

62. Evaluate $\int \frac{\sqrt[3]{x-8}}{x} dx$.